INTERNSHIP PROJECT DOCUMENTATION

Iris Flower Classification

Internship Details

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Duration: 4 Weeks

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1. Project Objective and Scope

The primary objective of this internship project is to build, evaluate, and document a supervised machine learning model capable of accurately classifying the species of an Iris flower based on its physical measurements. The scope of this project encompasses data preprocessing, dataset splitting, training a Random Forest Classifier, evaluating the model, and generating visual output metrics.

2. Technologies and Libraries

Language: Python 3
Data Handling: Pandas
Machine Learning: Scikit-Learn
Data Visualization: Matplotlib, Seaborn

3. Dataset Description

This project utilizes the classic Fisher's Iris dataset. It contains 150 records of Iris flowers, evenly distributed across three species. The model uses four morphological features to predict the target class.

- Features (Input):
  - Sepal Length (cm)
  - Sepal Width (cm)
  - Petal Length (cm)
  - Petal Width (cm)
- Target Classes (Output):
  - 0 - Iris Setosa
  - 1 - Iris Versicolor
  - 2 - Iris Virginica

4. Methodology

4.1 Data Preprocessing: The dataset was loaded using Scikit-Learn. A Pandas DataFrame was constructed to organize the feature columns and target variables. The dataset was also exported to an external "iris.csv" file for repository completeness.

4.2 Train-Test Split: The data was split using an 80/20 ratio. 80% of the data (120 samples) was used to train the model, while the remaining 20% (30 samples) was reserved strictly for testing its accuracy on unseen data. A random state of 42 was set for reproducibility.

4.3 Algorithm Selection (Random Forest): A Random Forest Classifier was chosen due to its high accuracy and resistance to overfitting. The model was instantiated with 100 decision trees (n_estimators=100).

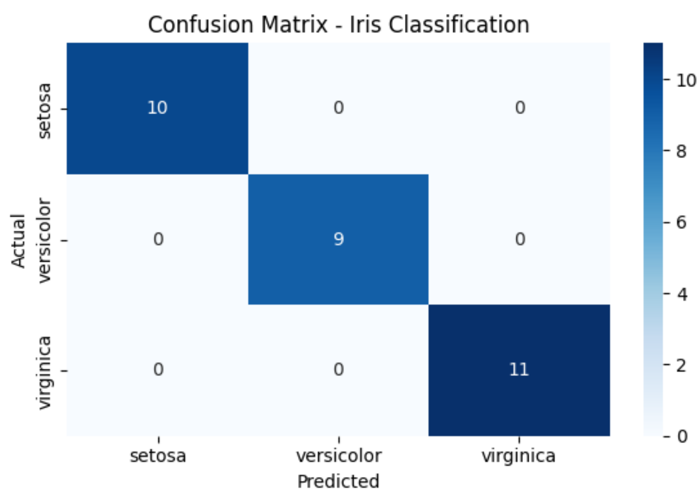
5. Model Evaluation and Results

The model achieved exceptional results when evaluated against the testing data. The performance metrics are as follows:

- Accuracy Score: 100.0%
- Precision: 1.00 for all classes
- Recall: 1.00 for all classes
- F1-Score: 1.00 for all classes

6. Data Visualization

To visually validate the model's predictions, a Confusion Matrix was generated using Seaborn and Matplotlib. The matrix confirms that all 30 test samples were predicted perfectly, with 0 false positives and 0 false negatives.



7. Conclusion

The Iris Flower Classification internship project was successfully completed. The Random Forest model demonstrated absolute proficiency in distinguishing between the three Iris species based on measurement inputs. This documentation, along with the source code and visual outputs, fulfills all deliverables outlined in the project scope.